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**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS)**
**REGULAR EXAMINATION - March 2026**
**Fourth SEMESTER**
**Discipline Specific Core Course (DSC) - MG4DSCCSC200,MG4DSCECC201**
**- OOPS CONCEPTS USING JAVA**
(2024 ADMISSION ONWARDS)

**Duration: 1 hour 30 minutes** **Maximum Marks : 50**

**Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

Very Short Answer Questions

Answer **all** questions

Each question carries **2** marks

1. Illustrate the use of arithmetic and relational operators with a simple Java example. [A]/[CO1]
Arithmetic operators perform mathematical operations like addition, subtraction, multiplication, and division. Relational operators compare two values and return a boolean result. Give an Example.
2. Explain how a base class constructor is accessed from a derived class constructor in Java and mention the keyword used for this purpose. [U]/[CO2]
A base class constructor is accessed from a derived class constructor using the super keyword. super() is used to call the constructor of the parent (base) class. It must be the first statement in the derived class constructor. Give an Example.
3. Name the Swing class used to create a tick box in Java. [K]/[CO3]
The Swing class used to create a tick box (checkbox) in Java is: JCheckBox
4. Identify and examine the purpose of an interface in Java. [An]/[CO2]
An interface in Java is used to define a contract that classes must implement. An interface in Java is a collection of abstract methods and constants. Purpose: Achieves abstraction (only method declarations, no implementation), Supports multiple inheritance.
5. State any two characteristics of a constructor. [K]/[CO2]
Two characteristics of a constructor are: It has the same name as the class, It does not have a return type (not even void).
Additional points (optional): It is called automatically when an object is created, It is used to initialize object variables.

**Part B**

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. What are labelled loops in Java? Explain their purpose with syntax. [U]/[CO1]
A labelled loop in Java is a loop that is assigned a name (label). It is mainly used with break and continue statements to control the flow of nested loops. Purpose: Helps to exit or continue an outer loop from within an inner loop. Improves control in complex nested loop structures. Give an example.
7. Illustrate various ways of initializing one-dimensional arrays and examine their differences.

[An]/[CO2]

A one-dimensional array in Java can be initialized in several ways:

1. Declaration and allocation separately: int arr[];

arr = new int[5];

2. Declaration and allocation together: int arr[] = new int[5];

3. Initialization with values: int arr[] = {1, 2, 3, 4, 5};

Differences: First two create arrays with default values (0). Last one assign explicit values at creation. {} syntax cannot be used after declaration separately.

8. Use the Delegation Event Model to explain how events are handled in Java. [A]/[CO3]

Java uses the Delegation Event Model for handling events in GUI applications.

Components: Event Source – Component that generates the event (e.g., button)

Event Object – Contains information about the event. Event Listener – Handles the event

Working: A source generates an event. The event is passed to a registered listener. The listener processes the event using handler methods. Give an Example

9. Explain Polymorphism. [U]/[CO1]

Polymorphism means "many forms". It allows one method or object to behave differently in different situations. Types: Compile-time polymorphism (Method Overloading), Run-time polymorphism (Method Overriding)

It improves code flexibility and reusability. Achieved using inheritance and interfaces.

Give an Example

10. Describe abstract classes and abstract methods in Java. [K]/[CO2]

Abstract Class: A class declared using abstract keyword. Cannot be instantiated. May contain abstract and non-abstract methods.

Abstract Method: A method declared without implementation. Must be implemented by subclasses. Give an Example:

11. Explain the role of javax.swing and java.applet packages in Java applications. [K]/[CO3]

javax.swing: Used to create Graphical User Interfaces (GUI). Provides lightweight components like: JButton, JFrame, JTextField

java.applet: Used to create applet-based applications. Applets run inside a web browser or applet viewer. Extends Applet class

Difference: Swing → Desktop GUI applications. Applet → Browser-based applications (now deprecated)

12. Identify the different types of inheritance supported by Java and illustrate each with suitable examples. [U]/[CO2]

Java supports the following types of inheritance:

1. Single Inheritance

```
class A {    void show() {}  }
class B extends A {}
```

2. Multilevel Inheritance

```
class A {}
class B extends A {}
class C extends B {}
```

3. Hierarchical Inheritance

```
class A {}
class B extends A {}
class C extends A {}
```

4. Multiple Inheritance (through interfaces only)

```
interface A {}
interface B {}
```

class C implements A, B {}
Java does not support multiple inheritance using classes. It is achieved using interfaces.

**Part C**

Essay Questions

Answer any **1** question

Each question carries **15** marks

13. Discuss the key features of Java in detail. [K]/[CO1]

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It has several powerful features:

1. Platform Independent: Java follows the principle "Write Once, Run Anywhere (WORA)". Java programs are compiled into bytecode, which runs on the Java Virtual Machine (JVM). Hence, Java programs can run on any system with JVM installed
2. Object-Oriented: Java is based on object-oriented principles such as: Encapsulation, Inheritance, Polymorphism, Abstraction. Promotes modular and reusable code
3. Simple: Easy to learn compared to C++. Removes complex features like: Pointers, Operator overloading, Uses automatic memory management
4. Secure: Provides strong security features: No pointer manipulation, Security manager, Suitable for network-based applications
5. Robust: Strong memory management. Exception handling mechanism. Eliminates runtime errors like memory leaks
6. Multithreaded: Supports execution of multiple threads simultaneously. Improves performance and efficiency
7. Distributed: Supports distributed computing using: RMI (Remote Method Invocation), Networking libraries
8. High Performance: Uses Just-In-Time (JIT) compiler, Converts bytecode into native machine code at runtime
9. Portable: Bytecode can be carried to any platform. Consistent behavior across systems
10. Dynamic: Classes can be loaded dynamically at runtime. Supports dynamic memory allocation

Java is widely used due to its portability, security, and robustness, making it suitable for web, desktop, and enterprise applications.

14. Explain how two methods can be defined in a class and how the user calls them from the main program. Illustrate with an example. [U]/[CO2]

In Java, methods are defined inside a class to perform specific tasks. These methods can be called from the main() method using an object.

Defining Methods:

Methods are defined using:

returnType methodName(parameters) { // body }

Calling Methods:

Methods are called using an object:

objectName.methodName();

Give an Example Program